

3-MONTH SONIA FUTURES (SO3)

Spread & Curve Structure — Fundamental Reference

The UK STIR Market — Bank of England Policy and the Sterling Rate Curve

This document presents a fact-based reference on the fundamental drivers of ICE Three Month SONIA Index Futures (commonly referred to as SO3, or simply "the SONIA contract"), traded on ICE Futures Europe in London, with particular focus on the forces that shape the forward rate curve and the spread structure between contract months. SONIA futures are the primary exchange-traded instrument for hedging and speculating on the path of UK short-term interest rates, having replaced Short Sterling (3-month GBP LIBOR) futures as the market standard following the LIBOR transition completed at the end of 2021. The SO3 curve is the most liquid and institutionally important sterling short-term interest rate (STIR) futures market, and the UK's structural counterpart to the SOFR market in the US (covered in a separate reference document in this series) and the EURIBOR/ESTR market in the Eurozone. Its spreads and flies encode the market's probability-weighted expectations about Bank of England monetary policy, UK inflation, growth, and financial conditions — across a horizon extending several years forward. Understanding the mechanics, the data, and the analytical framework for this market, including its structural similarities to and differences from the US SOFR market, is the subject of this document.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Settlement Mechanics

Three Month SONIA futures replaced Short Sterling (3-month GBP LIBOR) futures as the benchmark UK short-term interest rate contract on ICE Futures Europe following the permanent cessation of GBP LIBOR at the end of 2021. The contract structure closely mirrors the Short Sterling conventions it replaced, and is itself structurally very similar to the CME SR3 (SOFR) contract covered in the companion reference document in this series, since both were designed as part of the same global benchmark-rate reform process and price an analogous compounded overnight rate.

Parameter	Detail
Exchange	ICE Futures Europe (London)
Ticker	SO3 (front month); the contract is formally named "Three Month SONIA Index Futures"
Underlying	Compounded daily SONIA (Sterling Overnight Index Average) over the contract's 3-month accrual period, calculated via the published EDSP formula
Contract size	£500,000 notional, structured so that the contract has a standardised basis point value for hedging purposes (£12.50 per basis point for most expiries, £6.25 for the front delivery month in the period immediately preceding last trading day)
Quotation	100 minus the annualised 3-month SONIA rate. Example: a price of 96.42 implies a rate of 3.58%. As with SOFR futures, prices rise when rates fall and fall when rates rise — the same inverted quotation convention used throughout the global STIR futures market.
Minimum tick	0.005 (half a basis point) = £12.50 per contract for most delivery months; 0.0025 (£6.25) for the front delivery month from the first business day of the calendar month preceding last trading day
Contract months	March (H), June (M), September (U), December (Z) — quarterly cycle, named by the start date of the accrual period, such that 25 delivery months are available for trading at any time (approximately 6 years forward); shorter-dated serial monthly contracts are also listed for the nearest months.
Settlement type	Cash-settled. The Exchange Delivery Settlement Price (EDSP) is calculated by compounding the daily SONIA fixings over the contract's accrual period using a published formula, analogous in structure to SR3's settlement against compounded SOFR.
Last trading day	One business day prior to the third Wednesday of the next quarterly delivery month; trading ceases at 18:00 London Local Time.
Reference (accrual) period	The 3-month period beginning on the delivery month's start date. As with SOFR futures, the contract settles to the realised compounded rate over this period — not a single forward-looking fixing — meaning the final settlement value is only fully known once the accrual period has concluded.
Matching algorithm	Central order book uses a gradual time-based pro-rata (GTBPR) matching algorithm with time-weighting, giving priority to the first order at the best price, subject to a minimum order size (collar) and a maximum order size (cap) — a market-microstructure detail distinct from CME's pure FIFO matching for SR3.

1.1 SONIA: The Underlying Rate

The Sterling Overnight Index Average (SONIA) is administered by the Bank of England, which took over administration of the benchmark as part of the broader UK benchmark reform programme following the LIBOR scandal. Understanding what SONIA measures is essential for interpreting the futures curve:

- **What SONIA measures:** SONIA is calculated using actual transactions and committed quotes sourced from the unsecured overnight sterling deposit market, covering wholesale deposits transacted by banks and building societies, money market funds, and other financial institutions with eligible counterparties. Unlike SOFR, which is a **secured** rate (collateralised by US Treasury repo), SONIA is an **unsecured** overnight rate — a structural difference between the two benchmarks that traders comparing the SO3 and SR3 curves should bear in mind.
- **Daily publication:** SONIA is published by the Bank of England each business day at approximately 9:00 AM London time, representing the prior business day's rate — the same general next-day publication convention as SOFR's publication by the NY Fed.
- **SONIA vs GBP LIBOR:** GBP LIBOR was a forward-looking term rate, panel-bank-submitted rather than transaction-based, and was discontinued amid concerns about both its declining transaction volume base and historical manipulation. SONIA is overnight, transaction-based, and backward-looking — the compounded SONIA for a 3-month period is only known at the end of that period, the same structural feature described for SOFR in the companion US reference document. SO3 futures therefore price the expected compounded overnight rate over the reference quarter, not a single forward-looking fixing.
- **SONIA and quarter-end/year-end dynamics:** as with SOFR, SONIA is sensitive to balance-sheet pressures around UK bank reporting dates and to gilt settlement flows, creating observable seasonal patterns in the daily fixing around quarter-end and year-end — addressed in detail in Section 6.4.

1.2 Price Quotation and Spread Arithmetic

The inverted price quotation (100 minus rate) operates identically to the SOFR convention described in the companion reference document, and the same spread and fly arithmetic applies directly to the SO3 curve:

- **Spread = C1 price minus C2 price.** A positive spread means the front contract is priced higher than the back — i.e. UK rates are expected to FALL between the two reference quarters, corresponding to an inverted Bank Rate expectations path over that horizon.
- **Negative spread:** front contract priced lower than back — UK rates expected to RISE.
- **The fly:** long C1 and C3, short 2×C2, capturing the market's view of whether a specific Monetary Policy Committee (MPC) meeting — described in Section 2 — will deliver a different outcome than what adjacent meetings are pricing, directly analogous to the FOMC-meeting fly described for SR3.

2. Bank of England Policy — The Primary Driver

The Bank of England's Monetary Policy Committee (MPC) sets Bank Rate, which is the primary anchor for all UK short-term interest rates including SONIA. Understanding MPC mechanics is the prerequisite for any analytical framework applied to the SO3 curve.

2.1 Bank Rate and SONIA

- **Bank Rate:** the MPC sets a single rate (in contrast to the Fed's target range) for the rate of interest paid by the Bank of England on commercial bank reserves held at the Bank — the UK's primary monetary policy instrument since the Bank moved to its current "floor system" of monetary control.
- **The Bank of England's floor system:** under this framework, Bank Rate is paid directly on the reserves balances that commercial banks hold at the Bank of England, anchoring overnight market rates near Bank Rate without the Bank needing to actively manage the quantity of reserves to hit its target — a structurally similar concept to the Fed's IORB-based "ample reserves" framework described in the SR3 reference document, though implemented through the UK's own specific operational mechanics.
- **SONIA's relationship to Bank Rate:** SONIA trades in close proximity to Bank Rate in normal market conditions, since the unsecured overnight market is directly anchored by the rate the Bank pays on reserves. Unlike SOFR, which (being secured) trades near the bottom of the Fed Funds target range, SONIA — as an unsecured rate — tends to track Bank Rate itself very closely rather than sitting at a structural discount to it, reflecting the difference between the UK's single-rate floor system and the US's target-range-with-secured-rate framework.

2.2 MPC Meeting Schedule and Its Direct Impact on the SO3 Curve

The MPC meets eight times per year on a published schedule, mirroring the FOMC's eight-meetings-per-year cadence. Each meeting produces a policy decision (hold, hike, cut) and is accompanied by a published vote split among committee members — a degree of individual-member transparency that is somewhat more granular, on a per-meeting basis, than the FOMC's practice of reporting a vote tally without the same systematic real-time disaggregation):

- Each quarterly SO3 contract covers a 3-month reference period that typically contains one or two MPC meetings. The price of each contract reflects the expected compounded SONIA over that period — the expected average Bank Rate path for that quarter, the same "days in quarter" mechanical relationship described for SR3 and FOMC meetings in the companion reference document.
- **Published vote split:** each MPC decision is accompanied by the individual voting record of each of the nine committee members (nominally: the Governor, Deputy Governors, the Chief Economist, and external members appointed by HM Treasury), published simultaneously with the decision. A split vote (e.g. 6-3 in favour of a hold) is itself a market-moving signal about the distribution of views within the committee and the likelihood of a near-term shift, distinct in granularity from the FOMC's aggregate vote reporting.

2.3 The Monetary Policy Report (Formerly the Inflation Report)

- Published quarterly alongside four of the eight MPC decisions (historically referred to as "Super Thursday" before the Bank moved to its current communication calendar), the Monetary Policy Report contains the Bank's own GDP growth and inflation forecasts (presented as a probability-weighted fan chart rather than the FOMC's point-estimate "dot plot"), providing the most direct forward-looking signal of the MPC's own assessment of the economy and the implied rate path consistent with returning inflation to the 2% target over the forecast horizon.
- **The fan chart vs the dot plot:** a structural communication difference from the Fed worth noting explicitly: the MPC's fan chart presents a probability distribution of outcomes around a central

projection, conditioned on a specified market-implied path for Bank Rate, rather than the FOMC's practice of collecting and publishing individual members' specific point forecasts for the policy rate itself. This means the SO3 curve's relationship to the Bank's own published forecasts requires a slightly different interpretive framework than the direct dot-plot-to-SR3-curve mapping described for the Fed.

3. Macroeconomic Fundamentals — The Rate-Setting Inputs

The MPC's remit, set by the UK Government, is a single target: 2% CPI inflation (in contrast to the Fed's dual mandate of maximum employment and price stability), with the MPC required to balance this against output and employment volatility as a secondary consideration. The following economic variables have documented, systematic relationships with the SO3 curve.

3.1 Inflation Data

Inflation is, if anything, an even more singularly dominant variable for the SO3 curve than the equivalent data is for SR3, given the MPC's single inflation-target remit as opposed to the Fed's dual mandate.

Indicator	Publisher	Frequency	Release Timing	BoE Target / Reference	Curve Impact
CPI (headline)	ONS	Monthly	~2nd-3rd week of month, 07:00 London	2% — the MPC's sole statutory remit target	The single most important data release for SO3, given the MPC's single-target remit; immediate repricing across the curve
CPI (core, ex food/energy)	ONS	Monthly	Same release as headline CPI	Closely watched as the underlying inflation signal	Core surprises move the whole SO3 curve; particularly weighted in MPC commentary
CPIH (CPI including owner-occupier housing costs)	ONS	Monthly	Same release as CPI	ONS's preferred headline measure, though the 2% target itself is defined on CPI	A secondary cross-check measure; divergence from CPI itself is occasionally commented on by MPC members
Services inflation	ONS (CPI sub-component)	Monthly	Same release as CPI	The MPC's specific area of focus for assessing domestically-generated, persistent inflation, given services' lower exposure to global goods price swings	Hot services prints have, in recent UK cycles, been the single most cited reason for the MPC to delay or pause cuts
RPI (Retail Price Index)	ONS	Monthly	Same release as CPI	Not a policy target (officially designated a "legacy" measure by ONS) but still referenced for index-linked gilts and some commercial contracts	Limited direct SO3 relevance but watched by gilt market participants for linker-related basis effects
Pay growth / Average Weekly Earnings (AWE)	ONS	Monthly	~6 weeks after reference period, 07:00 London	The MPC's primary wage inflation gauge, explicitly referenced as a driver of services inflation persistence	Regular pay growth surprises move the SO3 curve materially, particularly in periods where the MPC has flagged wage growth as the binding constraint on cuts

3.2 Labour Market Data

The UK labour market data infrastructure has, in recent years, been subject to well-documented sample-size and response-rate challenges affecting the ONS Labour Force Survey, a UK-specific data-quality consideration that traders should weigh when assessing the reliability of headline labour market prints relative to the equivalent (generally more robust) US data infrastructure:

- **ONS Labour Market Overview (unemployment rate, employment rate, vacancies):** published monthly, combining the Labour Force Survey with PAYE/RTI administrative tax data. Following documented LFS sample-size issues, the ONS has increasingly emphasised the administrative PAYE data as a more reliable real-time signal, a UK-specific data-quality caveat with no direct parallel in the well-established US BLS survey infrastructure.
- **Vacancies-to-unemployment ratio:** the UK's equivalent labour-market-tightness gauge to the US JOLTS V/U ratio described in the SR3 reference document, tracked by the MPC as an indicator of how much slack exists in the labour market.
- **Average Weekly Earnings (AWE), regular pay (ex bonuses):** as referenced in Section 3.1, this is the MPC's primary wage data input, explicitly and repeatedly cited in MPC commentary as a key determinant of the pace of disinflation in services prices.
- **HMRC PAYE real-time information (RTI):** monthly, considered a more timely and robust payrolled-employee count than the LFS-based survey figures, increasingly cross-referenced by market participants alongside the headline LFS release given the sample-size concerns noted above.

3.3 Economic Activity and Growth Data

- **GDP:** published monthly by the ONS (a structural difference from the US's quarterly-only GDP release — the UK publishes a monthly GDP estimate in addition to the quarterly figure), giving UK market participants a higher-frequency activity signal than is available in the US data infrastructure.
- **S&P; Global/CIPS UK PMIs (Manufacturing, Services, Composite):** monthly, the UK's equivalent to the US ISM surveys described in the SR3 reference document, with the Composite PMI (combining manufacturing and the much larger services sector) the most closely watched single figure given services' dominant share of UK GDP.
- **Retail Sales:** monthly, ONS, a UK consumer demand signal analogous to the US Census Bureau release described for SR3.
- **GfK Consumer Confidence:** monthly, a UK-specific sentiment gauge tracked as a leading indicator of consumer spending intentions.

3.4 Financial Conditions and Gilt Market Indicators

- **UK gilt yields:** as with the US Treasury-SR3 relationship described in the companion document, the 2-year gilt yield is the closest cash-market equivalent to the front SO3 strip, and gilt market stress episodes (most notably the September–October 2022 "mini-budget" gilt crisis, discussed in Section 8.1) have historically had a more acute and direct feedback effect on UK monetary policy expectations than is typical of equivalent US Treasury market episodes, given the UK's relatively larger reliance on overseas gilt demand and the more concentrated domestic pension/LDI (liability-driven investment) sector exposure to long-dated gilt yields.

4. The Rate Cycle and SO3 Curve Regimes

As with SR3, the SO3 curve takes on fundamentally different shapes in different phases of the monetary policy cycle. The general regime framework described in the SR3 reference document applies directly to SO3; this section highlights the UK-specific considerations relevant to identifying and trading each regime.

Regime	Rate Direction	Curve Shape	UK-Specific Consideration
Hiking cycle	Rising rates	Inverted (front > back in price). Curve flattens as hikes priced in.	UK hiking cycles have, in recent history, been complicated by the relatively higher pass-through of import prices (given the UK's greater trade openness and energy import dependence) to headline CPI, meaning external/global cost shocks have a documented tendency to drive UK-specific hiking cycles independent of domestic demand conditions.
Peak / pause	On hold at terminal rate	Flat to mildly inverted near front; back curve may begin pricing cuts.	The MPC's explicit services-inflation and pay-growth focus (Section 3.1–3.2) means the pause phase is often extended relative to a pure headline-CPI-only framework, as the MPC has repeatedly signalled it wants to see services inflation and pay growth cool before committing to cuts.
Cutting cycle	Falling rates	Front falls in yield (rises in price); back anchored by long-run neutral rate expectations.	UK cutting cycles are watched closely for their effect on the mortgage market, given the UK's relatively higher share of short-dated fixed-rate (2- and 5-year) mortgage products compared to the US's dominant 30-year fixed structure — meaning UK rate cuts feed through to household disposable income on a faster and more visible refinancing cycle than in the US.
Post-cycle / normalisation	Approaching neutral	Flat or mildly upward sloping in yield terms.	UK neutral rate (r^*) estimates are subject to the same general uncertainty described for the US, compounded by UK-specific structural questions including post-Brexit trade and productivity trends and the long-run fiscal position (Section 9.2).

4.1 The Neutral Rate (R^*) — The UK-Specific Anchor

- The Bank of England, like the Fed, does not publish an official estimate of the UK neutral rate, though Bank staff research and external estimates (including UK-specific applications of Holston-Laubach-Williams-style models) are periodically referenced in MPC communications and speeches.
- UK-specific structural factors affecting r^* estimates:** post-Brexit trade friction and its effect on trend productivity growth, the UK's persistent productivity puzzle (weaker productivity growth than pre-2008 trend, a long-running and well-documented UK economic feature), demographic ageing, and the UK's fiscal position (Section 9.2) are all cited by analysts as inputs to the UK-specific neutral rate debate, distinct in their specific weighting from the US r^* discussion in the companion SR3

document though conceptually parallel.

4.2 The Term Premium and Gilt Supply

- As with the US Treasury market's effect on the SR3 back curve (via the ACM term premium model described in the companion document), UK gilt supply and the term premium embedded in long gilt yields affect the back end of the SO3 curve. The UK Debt Management Office (DMO) publishes the UK's gilt issuance remit, the UK-specific equivalent to the US Treasury's Quarterly Refunding Announcements described in Section 9.2 of the SR3 reference document.

5. Key Data Events and the SO3 Market Calendar

The SO3 curve responds to a well-defined calendar of UK data releases and Bank of England policy events, structurally parallel to (but distinct in specific publisher, timing, and emphasis from) the US calendar described in the SR3 reference document.

Event / Release	Publisher	Frequency	Timing	Primary Curve Impact	Magnitude
CPI (headline + core + services)	ONS	Monthly	~2nd-3rd week, 07:00	All curve — the single largest monthly event, given the MPC's single-target remit	★★★★★
MPC Rate Decision + Minutes	Bank of England	8× per year	12:00 London (decision + minutes published simultaneously)	All curve — immediate repricing; note minutes are released same-day with the decision, unlike the FOMC's 3-week-lag minutes	★★★★★
Monetary Policy Report (4× per year)	Bank of England	4× per year, with MPC decision	12:00 London	C2-C12 — fan-chart forecast revision; press conference follows at 12:30	★★★★★
Governor Press Conference	Bank of England	8× per year (all meetings since procedural changes)	~12:30 London	All curve — often moves more than the decision/minutes themselves	★★★★★
Average Weekly Earnings (pay growth)	ONS	Monthly	~6 wks after reference period, 07:00	C2-C8 — the MPC's primary wage inflation gauge	★★★★★
Labour Market Overview (unemployment, vacancies)	ONS	Monthly	Same release as AWE, 07:00	C1-C8 — subject to the sample-size caveats noted in Section 3.2	★★★
Monthly GDP estimate	ONS	Monthly	~6 wks after reference month, 07:00	C2-C8 — a higher-frequency UK-specific growth signal with no direct US equivalent	★★★
S&P; Global/CIPS Composite PMI	S&P; Global / CIPS	Monthly	~3rd-5th business day	C1-C4 — services-weighted activity leading indicator	★★★
Quarterly GDP (preliminary + revisions)	ONS	Quarterly	~Mid-month following quarter end	C2-C8 — confirms/revises the monthly estimates	★★★
Retail Sales	ONS	Monthly	~3rd week, 07:00	C1-C4 — consumer demand signal	★★
DMO gilt remit / issuance calendar updates	UK Debt Management Office	Per Budget/fiscal event	Variable	C8-C25 — gilt supply signal for the back end	★★★

Bank of England Decision Maker Panel (DMP) survey	Bank of England / Ipsos	Monthly	~Mid-month	C2-C6 — business-level inflation and wage expectations, an MPC-referenced survey input	★★
Bank of England Agents' summary / Citizens' Panel	Bank of England	Periodic, feeds into MPC decisions	Variable	Background — qualitative regional and household intelligence referenced in MPC discussion	★

5.1 The CPI Release — Anatomy of the Most Important UK Event

- As with the US CPI release's effect on SR3, UK CPI is consistently the single most market-moving scheduled event for the SO3 curve — arguably even more dominant than its US equivalent, given the MPC's single inflation-target remit compared to the Fed's dual mandate, which gives the Fed at least some institutional latitude to weight labour market data more heavily in ambiguous situations.
- **Services inflation as the MPC's specific focus:** as referenced in Section 3.1, the MPC has, in recent communication, repeatedly identified services inflation specifically (rather than the volatile goods/energy components more exposed to global commodity swings) as the key gauge of domestically generated, persistent inflation pressure — meaning a hot services print can move the SO3 curve's near-term pricing more than an equivalent headline surprise driven by energy or import prices.
- **Same-day minutes:** a structural difference from the Fed process worth re-emphasising — the MPC publishes its minutes simultaneously with the rate decision itself (both at 12:00 London time), rather than the FOMC's practice of releasing minutes three weeks after the decision. This means the SO3 curve receives the full detail of the policy debate, including the vote split, in a single release window, rather than the two-stage "decision-then-minutes" information release pattern that characterises the SR3/FOMC calendar.

6. Spread and Fly Mechanics on the SO3 Curve

6.1 What a Spread Expresses

An SO3 calendar spread operates identically in mechanism to the SR3 spread described in the companion reference document — it is a position on the change in the 3-month rate between two reference periods, not a position on the level of rates:

- **Positive spread (front > back in price):** the market implies UK rates will FALL between the front and back reference quarters. **Negative spread:** rates expected to RISE.
- **Spread size in rate terms:** a spread of 0.25 price points = 25bp of expected rate change between the two reference quarters — directly analogous to the SR3 calculation.
- **MPC meeting attribution:** each spread can be attributed to the MPC meetings falling in the gap between the two reference quarters, effectively pricing the probability-weighted expected rate change from those meetings.

6.2 What a Fly Expresses

- As with SR3, an SO3 fly captures curvature — the market's view of a temporary rate move centred on a specific contract's reference period, most commonly used to express uncertainty about whether a specific MPC meeting delivers a different outcome (hold vs cut, or a 25bp vs 50bp move) than what the meetings on either side of it are pricing.
- **UK-specific fly consideration:** because four of the MPC's eight meetings per year are accompanied by the quarterly Monetary Policy Report (Section 2.3) and four are not, flies centred on a "Report meeting" quarter can behave differently from flies centred on a "non-Report meeting" quarter, since the Report meetings carry additional forecast-revision information that non-Report meetings do not — a UK-specific structural asymmetry across the eight annual meetings with no precise equivalent in the FOMC's more uniform meeting cadence (though the FOMC's quarterly SEP/dot-plot meetings create a broadly analogous asymmetry).

6.3 Curve Zone Characteristics

Zone	Contracts (approx.)	Dominant Driver	Key Signals	Volatility
Ultra-front	C1-C2 (current quarter)	Known SONIA fixings, quarter-end dynamics, next MPC meeting	Daily BoE SONIA publication, gilt repo market levels	Low (mechanics-driven)
Front	C2-C4 (~6-12 months)	Near-term MPC path; CPI and pay growth surprises dominate	Monthly CPI, AWE pay data, MPC statement/minutes language, Governor speeches	Highest
Belly	C4-C8 (~12-24 months)	Terminal rate expectations; Monetary Policy Report fan chart; UK growth trajectory	Quarterly MPR, monthly GDP, Composite PMI	High
Mid-Back	C8-C25 (~24-72 months)	Neutral rate (R^*) estimates; gilt term premium; DMO issuance remit	DMO remit updates, fiscal events (Budget/Spring Statement), Bank staff R^* research	Moderate to Low

6.4 Quarter-End and Gilt Settlement Dynamics

SONIA, like SOFR, is affected by predictable, calendar-visible balance-sheet and settlement patterns, though operating through UK-specific market plumbing:

- **UK bank reporting date effects:** similar in concept to the US SLR-driven quarter-end repo tightening described for SOFR, UK bank balance-sheet management around regulatory reporting dates can create modest, predictable SONIA fixing patterns, though the UK's unsecured-overnight-deposit-based SONIA calculation methodology means the specific mechanics differ from SOFR's secured-repo-based calculation.
- **Gilt settlement and DMO auction calendar:** large gilt auction settlement dates, published in advance via the DMO's issuance calendar, can affect short-term sterling money market liquidity in a manner broadly analogous to the US Treasury settlement effects on SOFR described in the companion document, though operating through the UK's gilt market plumbing specifically.
- **Year-end dynamics:** UK money markets, like their US counterpart, can exhibit a year-end liquidity premium reflected in the December SO3 contract pricing, though the magnitude and specific drivers (UK bank regulatory reporting calendar, differing from the US SLR framework) are UK-specific and should not be assumed to mirror the US "turn premium" precisely.

7. Bank of England Balance Sheet — QE, QT, and Reserves

As with the Federal Reserve's balance sheet effect on SR3 described in the companion document, the Bank of England's balance sheet — expanded via Quantitative Easing (QE, branded by the Bank as the "Asset Purchase Facility," APF) since 2009 and now being unwound via Quantitative Tightening (QT) — affects the SO3 curve through reserve supply and term premium channels.

7.1 The Asset Purchase Facility (APF)

- The Bank's QE programme, conducted through the APF, involved gilt purchases (and a smaller corporate bond purchase scheme) funded by central bank reserve creation, structurally analogous to the Fed's QE programme described in the SR3 reference document, operating through UK gilts rather than US Treasuries and agency MBS.

7.2 Active Gilt Sales — A UK-Specific QT Feature

- **The key structural difference from the Fed's QT approach:** unlike the Fed, which has conducted QT purely through passive balance-sheet runoff (allowing maturing securities to roll off without reinvestment, as described in the SR3 reference document), the Bank of England has, at various points, supplemented passive runoff with **active gilt sales** from the APF portfolio — a more aggressive QT mechanism with no direct Fed parallel, reflecting the Bank's own judgement about the appropriate pace of balance sheet normalisation.
- **Indemnity arrangement with HM Treasury:** the APF operates under an indemnity from HM Treasury, meaning losses incurred on QE gilt holdings as they are sold or mature at a loss (a mechanical consequence of the Bank having purchased gilts at higher prices/lower yields than prevail in the current higher-rate environment) are borne by the UK Exchequer — creating a direct fiscal cost of QT that has, at various points, been the subject of public and political commentary in the UK in a way that has no precise equivalent in the US framework, where the Fed's own balance sheet losses are absorbed differently within the Fed's own accounting framework rather than creating an explicit, publicly tallied Treasury cost.

7.3 Reserves and the UK's Floor System

- As described in Section 2.1, the UK operates a floor system in which Bank Rate is paid on all reserves, a structurally simpler framework in some respects than the Fed's IORB-plus-ON-RRP dual-rate system, though both frameworks share the underlying "ample reserves" philosophy of controlling rates through administered rates rather than through active management of reserve scarcity.

8. Cross-Market Relationships

8.1 SO3 and the UK Gilt Curve

SO3 futures and the UK gilt yield curve are related in the same general way SR3 relates to the US Treasury curve, though the UK gilt market has its own distinct structural features:

- **2-year gilt yield vs SO3 front curve:** the closest cash-market equivalent to the front SO3 strip, analogous to the 2-year Treasury's relationship to SR3.
- **The September-October 2022 gilt crisis:** a documented, UK-specific episode with no precise US parallel, in which the announcement of a package of unfunded tax cuts (the September 2022 "mini-budget") triggered a sharp sell-off in long-dated gilts, driven substantially by forced selling from pension fund liability-driven investment (LDI) strategies facing collateral calls as yields rose — a feedback loop between the gilt market and the UK pension sector's structural use of leveraged LDI strategies that is a distinctly UK market-structure feature. The Bank of England intervened with a temporary, targeted gilt purchase programme to restore market functioning, an episode that materially affected SO3 pricing at the time given its direct relevance to financial stability considerations feeding into MPC policy assessment.
- **LDI and the pension sector as a UK-specific gilt market structural feature:** the scale of UK defined-benefit pension liabilities hedged via leveraged gilt-based LDI strategies is a distinctly UK phenomenon (reflecting the UK pension industry's specific liability-matching practices) that creates a feedback channel between gilt yield moves and forced gilt-market flows without a precise US analogue of comparable scale.

8.2 SO3 and Sterling (GBP)

- Higher UK short rates (lower SO3 prices) typically support GBP by increasing GBP-denominated asset returns relative to foreign alternatives, the same general mechanism described for SR3 and the US dollar, operating in the opposite currency direction.
- **UK twin deficit considerations:** the UK's current account and fiscal deficit position has, at various points, been cited by analysts as a structural factor affecting GBP's sensitivity to UK-US or UK-Eurozone rate differentials, with the UK sometimes characterised as more reliant on overseas capital inflows to fund its twin deficits than is the case for the US given the dollar's reserve-currency status — a structural asymmetry relevant to how GBP responds to UK-specific risk events.

8.3 SO3 and Equity/Credit Markets

- The general relationships described for SR3 and equity/credit markets in the companion document — risk-off episodes pulling forward rate-cut expectations, the "central bank put" concept and its limits during inflationary episodes — apply in broadly similar fashion to SO3, operating through the FTSE 100/250 and UK credit market equivalents rather than US benchmarks, though UK equity and credit market moves are also frequently driven by global rather than purely UK-specific factors given the London market's internationally-oriented listed company base (the FTSE 100 in particular derives a large share of constituent company revenue from outside the UK).

9. Global Context — the Fed-ECB-BoE Triangle and Fiscal Policy

SO3 futures price UK rates, but — as with SR3 — they do not exist in isolation. The rate decisions of the Federal Reserve and the European Central Bank, the UK's two most significant trading-partner and capital-flow-linked economies, create cross-currency differentials that affect GBP, UK financial conditions, and feed back into the SO3 curve.

9.1 The Fed-ECB-BoE Relative Timing Relationship

- UK monetary policy cycles have historically tended to track a position somewhere between the Fed's and the ECB's relative timing in recent hiking and cutting cycles, reflecting the UK economy's structural position as more exposed to the same global energy and goods price shocks affecting the Eurozone, while also sharing some labour market and services-inflation dynamics more comparable to the US — though this relative-timing relationship is a loose empirical pattern rather than a structural rule, and has varied across different cycles depending on the UK-specific shocks affecting any given episode (e.g., the 2022 gilt crisis described in Section 8.1, which was a UK-specific fiscal-policy shock unrelated to the Fed or ECB cycle).
- **EUR/GBP and USD/GBP as joint inputs:** because the UK trades extensively with both the US and the Eurozone, GBP's effective exchange rate (and the imported-inflation channel it creates) responds to both the Fed-BoE and ECB-BoE rate differentials jointly, rather than to a single dominant cross-currency relationship of the kind the US dollar has with any single counterpart currency.

9.2 UK Fiscal Policy and Gilt Supply

As with US fiscal deficits and Treasury supply's effect on the SR3 back end (described in the companion document), UK fiscal policy and gilt issuance affect the SO3 back curve through the term premium channel, with some UK-specific institutional features:

- **The Office for Budget Responsibility (OBR):** the UK's independent fiscal watchdog, broadly analogous in function to the US Congressional Budget Office (CBO) described in the SR3 reference document, publishing fiscal and economic forecasts alongside each UK Budget and Spring Statement. OBR forecast revisions that imply a larger or smaller structural deficit shift gilt supply expectations and, through the term premium channel, the SO3 back end.
- **The DMO's annual financing remit:** published at each Budget, setting out the planned gilt issuance programme for the coming fiscal year by maturity bucket — the UK-specific equivalent to the US Treasury's Quarterly Refunding Announcements, though set on an annual rather than quarterly cycle.
- **The September 2022 episode as the canonical UK fiscal-gilt-monetary-policy feedback example:** as described in Section 8.1, the mini-budget episode is the clearest documented instance of UK fiscal policy directly and acutely affecting gilt yields and, through the financial stability and inflation-expectations channels, MPC policy assessment — a more acute and directly observable fiscal-monetary feedback loop than has typically been observed in the equivalent US Treasury-Fed relationship over a comparable recent period.

10. SO3 vs SR3 — A Structural Comparison

Because SO3 and SR3 were both designed as part of the same global IBOR-to-overnight-rate transition and share a very similar contract architecture, this section catalogues the specific structural differences a trader moving between the two markets should hold in mind, presented as a factual comparison only.

Dimension	SO3 (UK SONIA)	SR3 (US SOFR)
Exchange	ICE Futures Europe (London)	CME Group (Chicago)
Underlying rate type	Unsecured overnight deposit rate	Secured overnight repo rate
Rate administrator	Bank of England	Federal Reserve Bank of New York
Central bank policy instrument	Bank Rate (single rate, floor system)	Fed Funds target range (IORB + ON RRP)
Policy committee	MPC — 9 members, published individual vote split	FOMC — 12 voting members, aggregate vote tally
Meetings per year	8	8
Minutes timing	Same day as decision	3-week lag after decision
Forecast communication	Quarterly Monetary Policy Report — probability fan chart	Quarterly SEP — point-estimate "dot plot"
Contract notional	£500,000	\$1,000,000 (90/360 convention)
Tick value	£12.50 (£6.25 front month near expiry)	\$12.50 (half-bp); \$6.25 variant for front contracts in certain conditions
Listed horizon	~6 years (25 quarterly delivery months)	~10 years (40 quarterly delivery months)
QT mechanism	Passive runoff + active gilt sales	Passive runoff only
2022 stress episode	Gilt crisis (fiscal-policy-triggered, LDI-driven)	No directly equivalent 2022 episode; March 2020 repo stress is the closer historical US parallel

10.1 Why the Differences Matter for Cross-Market Positioning

- A trader running a relative-value position across SO3 and SR3 is, by construction, also taking a GBP/USD view, since comparing the two curves in common terms requires currency conversion — the same general principle described for cross-currency commodity pairs (e.g., the CC/XC cocoa and No.11/XW sugar comparisons) elsewhere in this document series, here applied to a rates rather than commodity context.
- The secured-vs-unsecured distinction between SOFR and SONIA means the two rates do not sit in analogous positions relative to their respective central bank policy rates — SOFR trades near the bottom of the Fed's target range, while SONIA tracks Bank Rate itself closely — a structural fact that should inform any basis calculation between the two curves.
- The same-day-minutes (UK) versus three-week-lag-minutes (US) difference means the information-arrival pattern around each central bank's decision day is structurally different: the SO3 curve receives the full committee debate detail immediately, while the SR3 curve receives an initial signal at the decision/press-conference stage and a second, separate information event three weeks later at the minutes release.

11. Summary Data Release Calendar

Release	Publisher	Frequency	Typical Timing	Primary SO3 Curve Impact
CPI (headline + core + services)	ONS	Monthly	~2nd-3rd week, 07:00	All curve — largest single monthly event given the MPC's single-target remit
MPC Rate Decision + Minutes	Bank of England	8× per year	12:00 London	All curve — immediate repricing; full committee debate detail released same-day
Monetary Policy Report	Bank of England	4× per year, with MPC decision	12:00 London	C2-C12 — fan-chart forecast revision
Governor Press Conference	Bank of England	8× per year	~12:30 London	All curve — often the largest single intraday move
Average Weekly Earnings	ONS	Monthly	~6 wks after reference period, 07:00	C2-C8 — primary wage inflation gauge
Labour Market Overview	ONS	Monthly	Same release as AWE, 07:00	C1-C8 — subject to LFS sample-size caveats
Monthly GDP estimate	ONS	Monthly	~6 wks after reference month, 07:00	C2-C8 — higher-frequency UK-specific growth signal
S&P; Global/CIPS Composite PMI	S&P; Global / CIPS	Monthly	~3rd-5th business day	C1-C4 — services-weighted activity signal
Quarterly GDP	ONS	Quarterly	~Mid-month following quarter end	C2-C8 — confirms/revises monthly estimates
UK Budget / Spring Statement + OBR forecast	HM Treasury / OBR	1-2× per year	Variable, announced in advance	C8-C25 — fiscal and gilt supply outlook; back-end term premium
DMO annual financing remit	UK Debt Management Office	Annual, at Budget	Variable	C8-C25 — gilt issuance programme by maturity bucket
BoE Decision Maker Panel survey	Bank of England / Ipsos	Monthly	~Mid-month	C2-C6 — business inflation/wage expectations
Bank of England daily SONIA publication	Bank of England	Daily	~09:00 London	C1 — daily fixing; front contract basis and convergence
HMRC PAYE RTI employment data	HMRC / ONS	Monthly	With Labour Market Overview	C1-C4 — cross-check against LFS-based employment figures

Three Month SONIA futures are, structurally, the United Kingdom's direct counterpart to the US SR3/SOFR market: both were built through the same global benchmark-reform process, price an analogous compounded overnight rate, and respond to a parallel central-bank-meeting-driven calendar. The material differences are not in the basic mechanics — the inverted price quotation, the spread and fly arithmetic, and the "days in quarter" attribution logic all transfer directly from SR3 to SO3 — but in

the specific institutional architecture each market sits within: an unsecured rather than secured underlying rate, a single inflation-target remit rather than a dual mandate, same-day rather than lagged committee minutes, and a domestic gilt market with its own distinct structural vulnerabilities, most vividly demonstrated by the September 2022 crisis.

For spread traders, the most reliable framework treats SO3 as governed by the same general analytical structure as SR3 — identify the current rate-cycle regime, attribute spreads and flies to specific MPC meetings using the "days in quarter" mechanics, and track the CPI/wage-growth/Monetary Policy Report calendar as the primary repricing triggers — while remaining alert to the UK-specific institutional features (the single-target remit's effect on the MPC's reaction function, the gilt market's LDI-driven vulnerability, and active gilt sales as a QT tool) that mean SO3 cannot simply be treated as a currency-converted copy of SR3.